How to build better systems? Value-centred approaches

Ethics and Bias in AI : Lecture 4

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So far…

- Developed knowledge of some examples of how AI impacts on people, and how it can lead to discrimination
- Have an understanding of key concepts relating to ethical AI (e.g. bias, fairness)
- Understand that there are a range of technically focused and human focused approaches to mitigating against harms

Up next…

Ethical approaches for the development of AI systems

→ Approaches to identify and evaluate harms in the development of AI systems.
→ Approaches to minmise harm and mitigate the effects of bias.

- Design based approaches
- Technical approaches
- Evaluation (e.g. Audits)
- Transparency (e.g. documentation, explainability)

→ Future trends & ethical frontiers

Lecture overview

- Values & technology
- Value Sensitive Design (VSD)
  - Introduce the VSD framework & tripartite structure
  - Stakeholders
  - Value tensions
- Application of VSD to AI/ML

What do we mean by value(s)?

1. Value as the quality of being useful and important: a perspective commonly used in design
2. <u>Value as what is important for people in life: a perspective less often considered in design</u>

Values

“a value refers to what a person or group of people consider important in life. In this sense, people find many things of value, both lofty and mundane: their children, friendship, morning tea, education, art, a walk in the woods, nice manners, good science, a wise leader, clean air.”

Friedman, Khan & Borning (2009)

Why are values important in design?

- Values are part of technical systems and devices; they are implicated in products and systems.

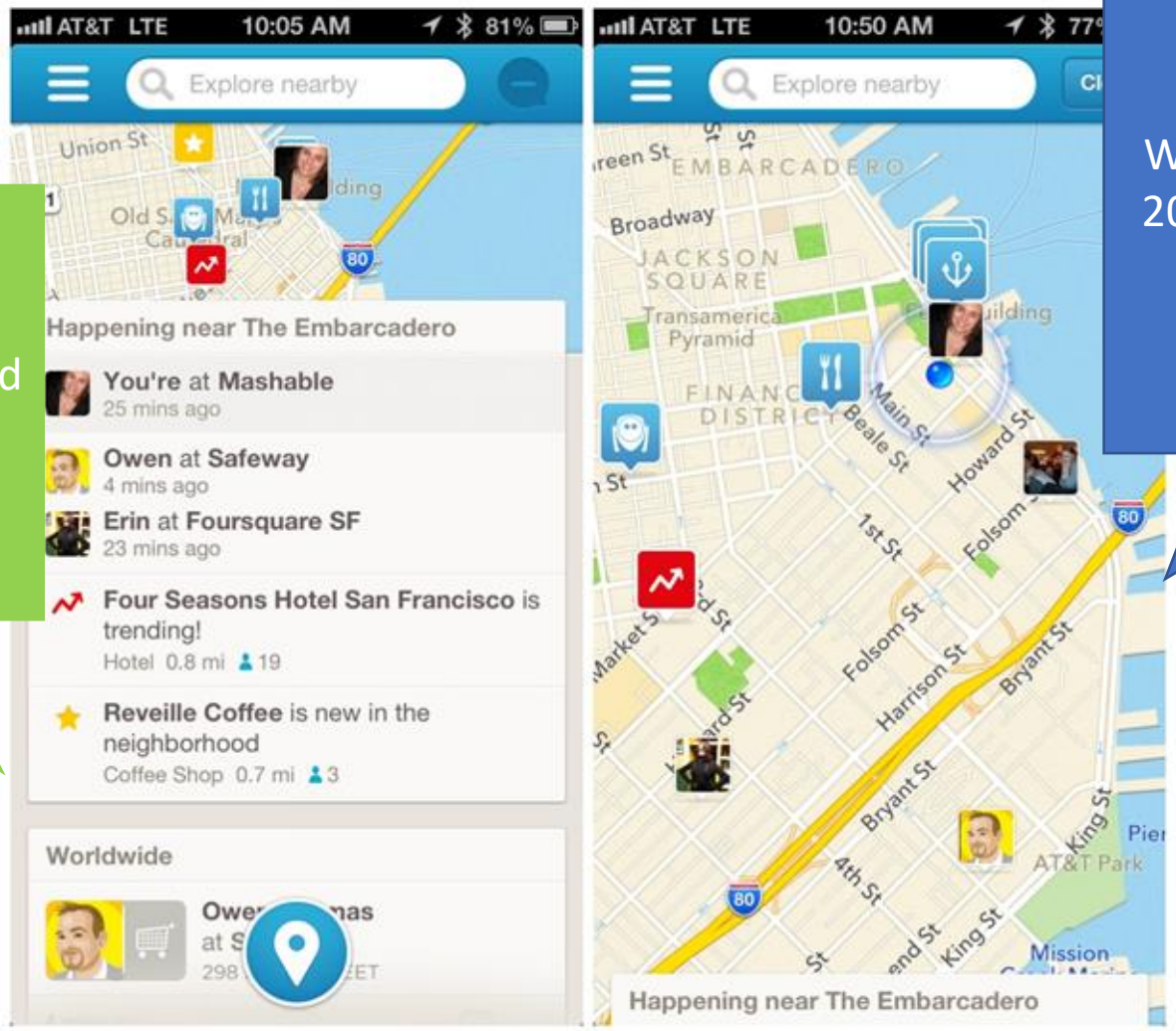
Typical example by Winner (1980) of low bridges ‘designed’ by Robert Moses to deny low-income people to travel to the beaches by bus.

What values are relevant to the design of a web browser?

Firefox GNU General Public License	**Microsoft Edge** Proprietary software	**Google Chrome** Freeware
Opera Freeware	**Internet Explorer** Proprietary software	**Safari** Freeware
Brave Mozilla Public License	**Chromium** BSD licenses	**Vivaldi** BSD licenses
Netscape Navigator Freeware	**Maxthon** Freeware	**Mosaic** Proprietary software

Technologies can lead to value tensions

Wonderful to know that my friend Helen is here as well



Why did you check in there at 20:00 yesterday? You said you were ill!

Firefox GNU General Public License	**Microsoft Edge** Proprietary software	**Google Chrome** Freeware
Opera Freeware	**Internet Explorer** Proprietary software	**Safari** Freeware
Brave Mozilla Public License	**Chromium** BSD licenses	**Vivaldi** BSD licenses
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Values

E.g.

- Efficiency
- Security
- Identity
- Calmness
- Informed consent
- Equality
- Privacy
- Trust
- Accountability
- Freedom from harm
- Freedom from bias
- Wellbeing
- Justice
- Autonomy
- Safety
- Sustainability
- Democracy

How can we address ethics/values in design?

- The notion of *user-centered design* (UCD) or *human-centered design* already implies some ethics. In user-centered design, both the design product and the design process aim to satisfy the needs and desires of the “user”.
- Value-Centered Design
- Participatory Design (PD) and Values-Led Participatory Design
- Value-Sensitive Design is one of the most reviewed approaches pertaining to values in technology design so far.

Value Sensitive Design

Value Sensitive Design

Value-Sensitive Design (VSD) acknowledges that technology design bears “directly and systematically on the realization, or suppression, of particular configurations of social, ethical, and political values”

(Flanagan et al. 2008).

Value-Sensitive Design

- Value-Sensitive Design is an attempt to systematically include values of ethical importance in design.
- VSD emerged in the 1990s from the insights of “human-computer interaction studies”.
- The traditional focus of engineers is on functionality, usability, efficiency, reliability, and affordability of (new) technologies.
- However: as we have seen, technology is not value-neutral, but has moral (and political) impacts on humans and their environment.
- VSD is oriented toward influencing the design of technology early in and throughout the design process

Value Sensitive Design: Key Principles

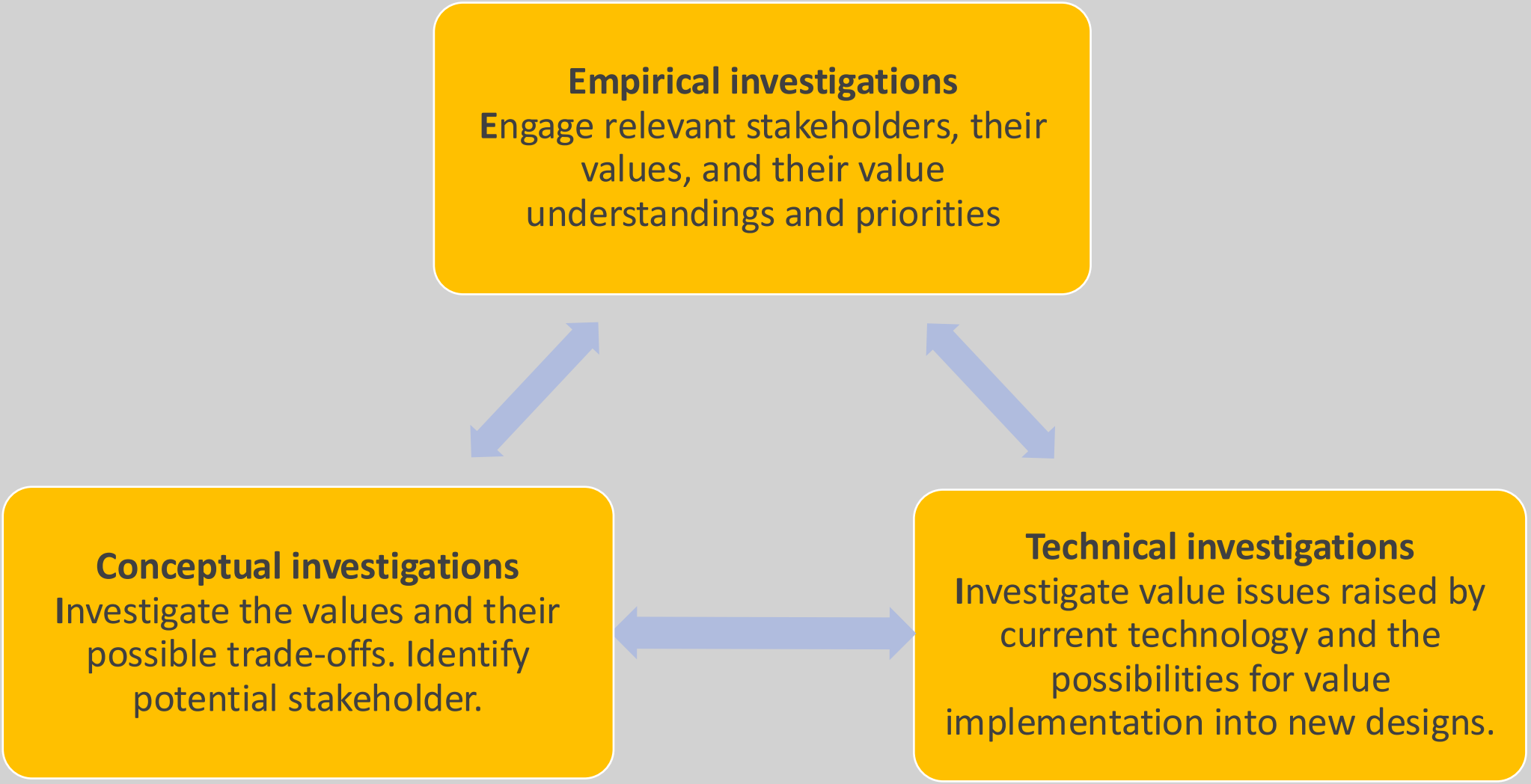
- Technologies are not neutral. Values can be expressed and embedded in technology.
- Values and choices made by some will have real effects (may not be obvious) on those who are affected (directly/indirectly).
  - Digital technologies (e.g. search engines, content moderation) have real impact on how people act and think.
- Conscious and explicit thinking about the values that are embedded in systems is morally significant.

Aim: to make the particular values at play explicit & evaluate how their implementations works out in practice & adjust accordingly

Value Sensitive Design: an interactional theory

- Values are viewed neither as inscribed into technology nor as simply transmitted by social forces
- A technology's actual use depends on the goals of the people interacting with it.
  - E.g. screwdriver can turn screws but it can also be used as a poker, but functions poorly as a ladle
- Therefore, need to consider potential use cases, but may not be able to predict all

VSD: Tripartite Structure



CONCEPTUAL INVESTIGATIONS

Conceptual analysis focuses on **identifying** and **defining** the relevant **values**, **stakeholders**, and **ethical considerations** associated with a technology.

- Comprise analytical, theoretical, or philosophically informed explorations of the central issues and constructs under investigation.
- VSD projects commonly develop careful (philosophically informed) working conceptualisations of specific values.
- Conceptual investigations are often generative, leading to framing propositions (design guidelines)

CONCEPTUAL INVESTIGATIONS

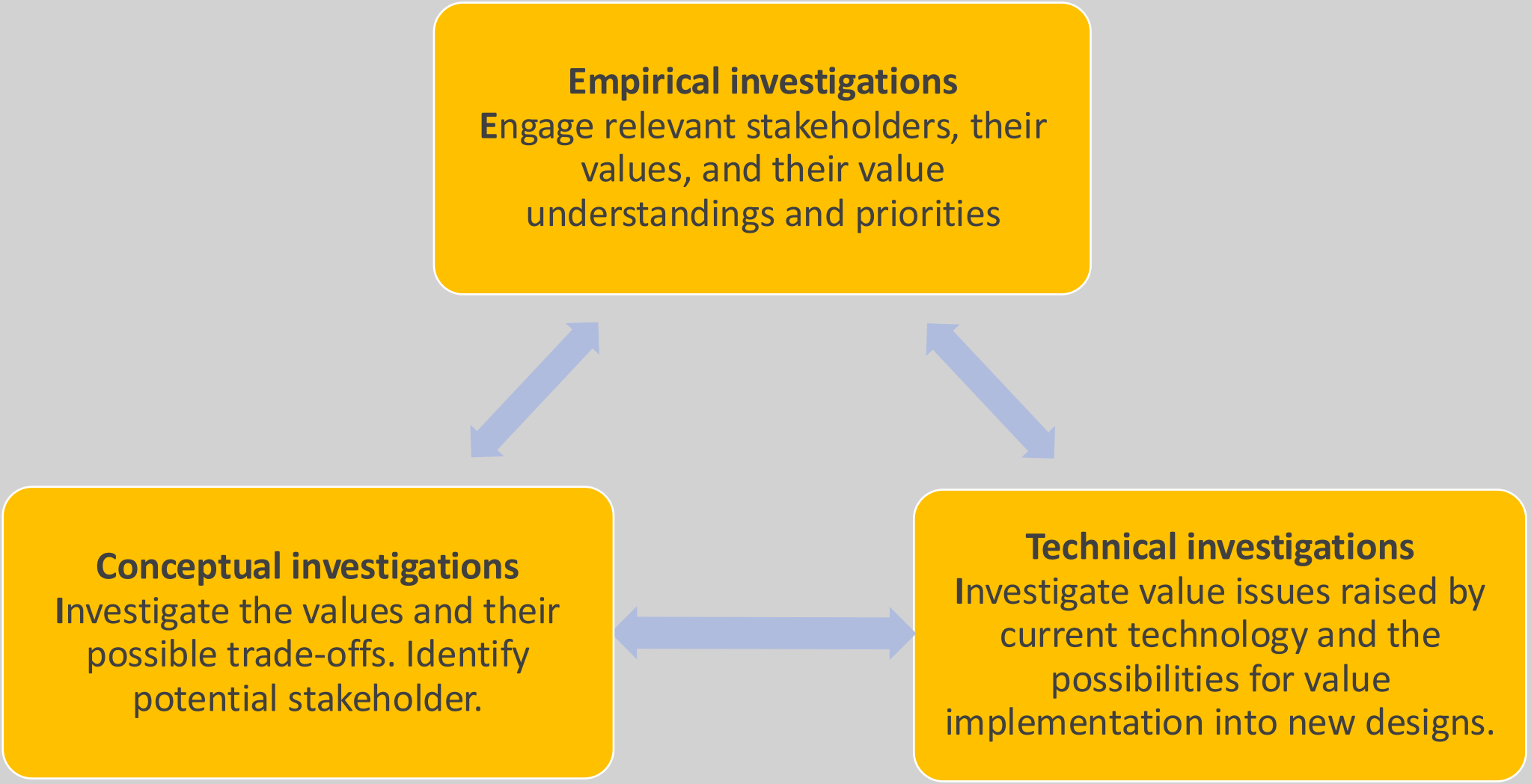
Questions like…

- What theoretical commitments and choice of conceptual framework are made?
- If the design team makes a commitment to particular ethical or cultural frameworks to support principled reasoning, how would it be articulated and integrated into the design process?
- What values are likely to be implicated and how can they be understood and characterised?
- What conceptual models for operationalising the value(s) will be employed?

Conceptual Investigation Example

- Identifying **values**:
  - Privacy, security, transparency, accountability, fairness.
- Defining **stakeholders**:
  - Citizens, government agencies, law enforcement, business owners, and civil rights organizations.
- Addressing ethical questions:
  - How can privacy be preserved while ensuring public safety?
  - Should individuals be notified when they are being monitored?
  - What mechanisms can prevent misuse of surveillance data?

VSD: Tripartite Structure



TECHNICAL INVESTIGATIONS

Technical analysis in VSD **evaluates** how **existing** or **potential technologies** support or undermine human **values**. It involves assessing system architecture, algorithms, and technical trade-offs.

- A given technology, game system or controller is more suitable for certain activities and more readily supports certain values, while rendering other activities and values more difficult to realise.
- These investigations have technology, system, controller as the unit of analysis
- Technical investigations exist in the spectrum between retrospective analysis of existing systems and proactive design of new systems to (better) support values identified in the conceptual and empirical investigations

TECHNICAL INVESTIGATIONS

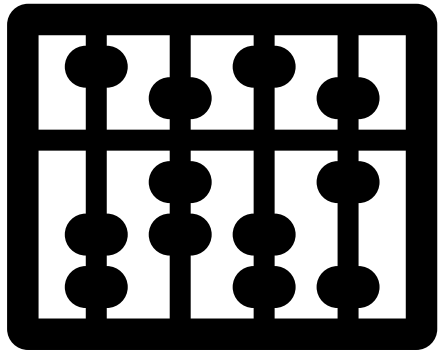
Questions like…

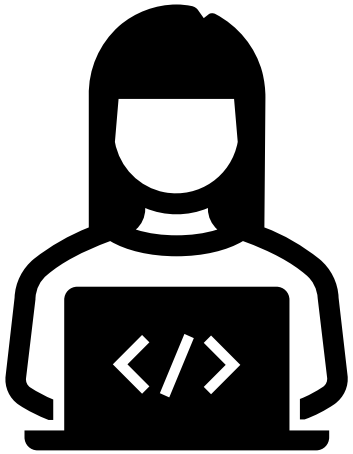
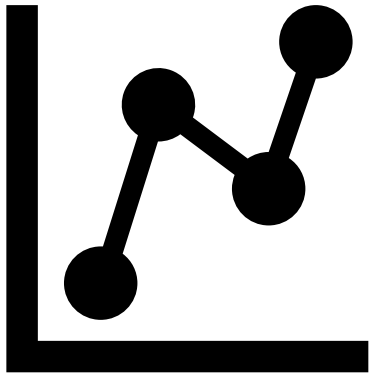
- What features of technical infrastructure enable, hinder, or even foreclose certain kinds of designs for supporting human activity?

- How do policies, laws, regulations create opportunities for technological development?

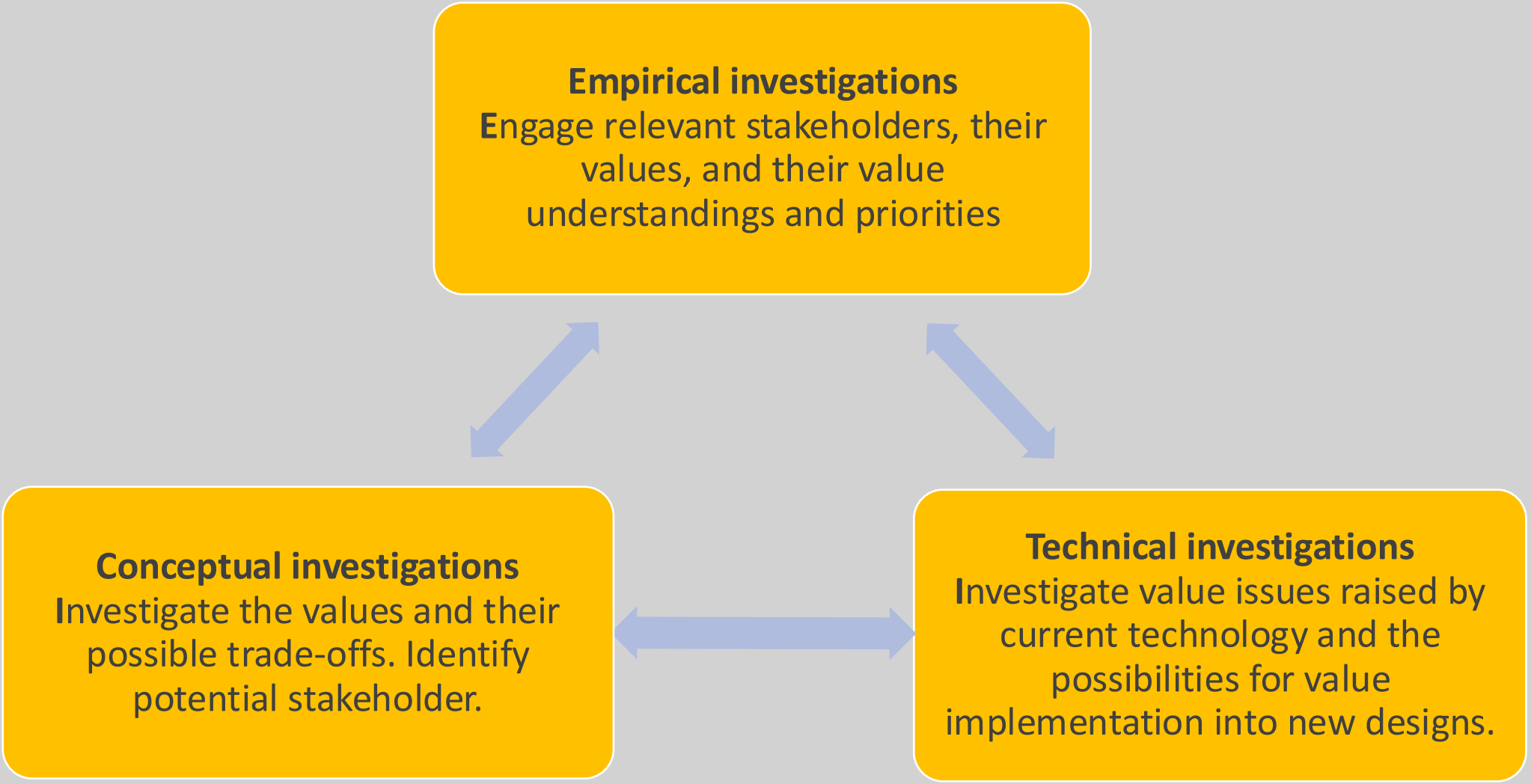
Technical Investigations Example

- To evaluate how existing or potential technologies support or undermine human values…
- Examining bias in training data
- Evaluating explainability
- Assess / test privacy safeguards and encryption methods to protect personal data
- Use fairness metrics to assess whether it disadvantages particular groups



VSD: Tripartite Structure



EMPIRICAL INVESTIGATIONS

Empirical investigations **validate** conceptual ideas by testing how people interact with technology in the real world. They also **inform technical decisions** by identifying usability issues and refining features based on user needs.

- Many analyses will need to be informed by empirical investigations of the human context in which the technology is situated drawing on both qualitative and quantitative methods.
- VSD projects are challenged to select a method that fits the particular project at a particular point in the design process.
- Empirical investigations in VSD projects can employ some of the 17 methods that have been appropriated to fit VSD

EMPIRICAL INVESTIGATIONS

Questions like:

- How do stakeholders apprehend values in the sociotechnical context?
- How do stakeholders prioritise competing values or otherwise envision resolution of value tensions?
- Are there difference between espoused practice (what people say) compared with actual practice (what people do)?

Empirical Investigation Example

- To understand how users feel about privacy, the team conducts empirical research..

For example…

- Surveys: Ask users about their concerns regarding voice data collection.
- Interviews: Speak with families, security experts, and privacy advocates.
- Observational Studies: Monitor how people use voice assistants in real-world settings.
- Analysis of usage data, review data of existing products

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Value sensitive design: Process

1. Start with a value, technology or context of use
2. Identify direct and indirect stakeholders
3. Identify benefits and harms for each Stakeholder group
4. Map benefits and harms onto corresponding values
5. Conduct an investigation of key values
6. Identify potential value conflicts
7. Integrate value considerations into one’s organizational structure

Friedman, Kahn & Borning (2009)

Whose values?

- Those who are or will be significantly implicated by the technology = **the stakeholders**
- STAKEHOLDER: refers to those people or entities who are or will be significantly implicated by a technology. They can be people, groups, neighborhoods, communities, organisations, institutions, or societies.
- WHY THESE STAKEHOLDERS?: VSD asks designers to seek out a robust set of stakeholders groups and to legitimate those stakeholders who have a good deal at stake - that is, to provide an analytical or empirical rationale for their inclusion in the design process

Stakeholders

Direct stakeholders: Individuals or organizations who interact with a system and its outputs. Often who commissioned it.

Indirect stakeholders: All other parties who are affected by the use of the system (i.e those who don't interact with the system but still impacted by it).

→ **Roles not individuals:** Defining stakeholders in terms of roles positions designers to understand how the same 'entity' could hold multiple relationships to the same technology

Stakeholders

- EXPLICITLY SUPPORTED PROJECT VALUES, DESIGNER VALUES AND STAKEHOLDER VALUES: Designers bring their own personal and professional values to their work. These are referred to as designer values. In the case of tensions between designer values, project values and stakeholder values, these tensions need to be mitigated = value workshop for both designers & stakeholders

Reading on Value Sensitive Design

- Friedman, B. and Hendry, D.G. (2019) Value sensitive design : shaping technology with moral imagination. Cambridge, Massachusetts: MIT Press.
- Friedman, B., Kahn, P. H., & Borning, A. (2009). Value Sensitive Design and Information Systems. In The Handbook of Information and Computer Ethics (1st ed., pp. 69–101). Wiley.
- Umbrello, S., & van de Poel, I. (2021). Mapping value sensitive design onto AI for social good principles. AI and Ethics, 1(3), 283-296.

Reading on Value Sensitive Design

VSD website: https://vsdesign.org/

Further reading

- Simon, J., Wong, P. H., & Rieder, G. (2020). Algorithmic bias and the Value Sensitive Design approach. Internet Policy Review, 9(4), 1-16. https://policyreview.info/pdf/policyreview-2020-4-1534.pdf
- de Reuver, M., van Wynsberghe, A., Janssen, M. *et al.* Digital platforms and responsible innovation: expanding value sensitive design to overcome ontological uncertainty. *Ethics Inf Technol* 22, 257–267 (2020). https://doi.org/10.1007/s10676-020-09537-z